Ahmed Heakl

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Research interests: efficient and verifiable vision-language models.

Education

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Aug 2026 – present
PhD in Computer Vision, expected May 2030 — *Abu Dhabi, UAE*
Advisors: Prof. Salman Khan and Prof. Fahad Shahbaz Khan.

Mohamed bin Zayed University of Artificial Intelligence (MBZUAI) — Aug 2024 – May 2026
MSc in Computer Vision, GPA 3.77/4.0, Outstanding Research Award — *Abu Dhabi, UAE*

Egypt-Japan University of Science and Technology (E-JUST) — Sep 2019 – Jul 2024
BSc in Computer Science and Engineering, GPA 3.85/4.0, ranked 1st of 56 — *Alexandria, Egypt*

Experience

Student Researcher | **Apertix** — Oct 2026 – Apr 2027
Abu Dhabi, UAE

- Building efficient streaming VLMs through recursive on-policy distillation, cutting latency by **4.2**× and raising performance on proactive benchmarks by **6%**.

Research Scientist Intern | **LG AI Research** — May 2026 – Oct 2026
Mentors: Sungik Choi and Moontae Lee — *Seoul, South Korea*

- Led **VQS**, program-verified self-evolution for VLMs, raising label correctness from 76% to **94%** and Qwen3-VL scores by up to **+3.18** points on 10 benchmarks.

Research Scientist Intern | **NAVER** — Jun 2025 – May 2026
Mentors: Martin Gubri, Sangdoo Yun, and Seong Joon Oh — *Tübingen, Germany*

- Led **Dr.LLM** (ICLR 2026), per-layer routers that let frozen LLMs skip or repeat blocks, giving **+3.4%p** accuracy with 3–11 fewer layers per query.

Research Associate | **MBZUAI** — Jun 2024 – May 2026
Advisors: Prof. Martin Takáč and Prof. Abdulrahman Mahmoud — *Abu Dhabi, UAE*

- Built SVRPBench (NeurIPS 2025 D&B) and the LLM transpilers CASS (ACL 2026) and GG (EMNLP 2025, **99.4%** x86-to-ARM accuracy).

Applied Scientist (remote) | **Monta AI** — Jul 2024 – Apr 2025
California, USA

- Trained a 7B VLM that turns charts and diagrams into code on 8 GPUs, co-led Arabic OCR over 100k documents (**95.3%** mAP@0.5), and built KITAB-Bench (ACL 2025).

Earlier: Software Engineer, And Africa Co., Ltd., Tokyo (remote, Mar 2023 – Jul 2024). Machine Learning Intern, Scoville Co., Ltd., Tokyo (Jan – Feb 2023).

Publications

* equal contribution. Full list on Google Scholar.

- **Program-Verified Self-Evolution for Vision-Language Models (VQS)** — Under review
  Ahmed Heakl, Sungik Choi, Moontae Lee, Salman Khan [Paper] [Project] [Code]
- **Dr.LLM: Dynamic Layer Routing in LLMs** — ICLR 2026
  Ahmed Heakl, Martin Gubri, Salman Khan, Sangdoo Yun, Seong Joon Oh [Paper] [Code]
- **The Geometry of Forgetting: Temporal Knowledge Drift as an Independent Axis in LLM Representations** — NeurIPS 2026
  Rania Elbadry*, **Ahmed Heakl***, Fan Zhang, Dani Bouch, …, Zhuohan Xie [Paper]
- **DocAtlas: Multilingual Document Understanding Across 80+ Languages** — EMNLP 2026
  Ahmed Heakl, Youssef Mohamed, Abdullah Sohail, Rania Elbadry, …, Salman Khan [Paper] [Project] [Code]
- **WorldCache: Content-Aware Caching for Accelerated Video World Models** — ECCV 2026
  Umair Nawaz, **Ahmed Heakl**, Ufaq Khan, Abdelrahman Shaker, Salman Khan, Fahad Shahbaz Khan [Paper] [Code]
- **Mobile-O: Unified Multimodal Understanding and Generation on Mobile Devices** — Under review
  Abdelrahman Shaker*, **Ahmed Heakl***, Jaseel Muhammad, Ritesh Thawkar, …, Fahad Shahbaz Khan [Paper] [Project] [Code]
- **VideoMolmo: Spatio-Temporal Grounding Meets Pointing**
  Ghazi Shazan Ahmad*, **Ahmed Heakl***, Hanan Gani, Abdelrahman Shaker, …, Salman Khan [Paper] [Project] [Code]

- **CASS: Nvidia to AMD Transpilation with Data, Models, and Benchmark** ACL 2026
  Ahmed Heakl, Gustavo Bertolo Stahl, Sarim Hashmi, …, Salman Khan, Abdulrahman Mahmoud [Paper] [Code]
- **LlamaV-o1: Rethinking Step-by-step Visual Reasoning in LLMs** ACL 2025 Findings
  Omkar Thawakar*, Dinura Dissanayake*, Ketan More*, Ritesh Thawkar*, **Ahmed Heakl***, …, Salman Khan [Paper] [Code]
- **KITAB-Bench: Multi-Domain Arabic OCR and Document Understanding Bench** ACL 2025 Findings
  Ahmed Heakl, Abdullah Sohail, Mukul Ranjan, Rania Hossam, …, Salman Khan [Paper] [Project] [Code]
- **GG: CISC-to-RISC Transpilation with Testing Guarantees** EMNLP 2025 Findings
  Ahmed Heakl, Sarim Hashmi, Chaimaa Abi, Celine Lee, Abdulrahman Mahmoud [Paper] [Project] [Code]
- **SVRPBench: A Realistic Benchmark for the Stochastic Vehicle Routing Problem** NeurIPS 2025 D&B
  Ahmed Heakl, Yahia Salaheldin Shaaban, Martin Takáč, Salem Lahlou, Zangir Iklassov [Paper] [Code]
- **CAMEL-Bench: A Comprehensive Arabic Large Multimodal Model Benchmark** NAACL 2025 Findings
  Sara Ghaboura*, **Ahmed Heakl***, Omkar Thawakar, Ali Alharthi, …, Rao M. Anwer [Paper] [Project] [Code]
- **AIN: The Arabic INclusive Large Multimodal Model** arXiv preprint
  Ahmed Heakl, Sara Ghaboura, Omkar Thawkar, Fahad Shahbaz Khan, …, Salman Khan [Paper] [Project] [Code]
- **SAHM: A Benchmark for Arabic Financial and Shari'ah-Compliant Reasoning** ACL 2026 Oral
  Rania Elbadry, Sarfraz Ahmad, **Ahmed Heakl**, Dani Bouch, …, Zhuohan Xie [Paper] [Project] [Code]
- **MedRoundsQA: Persona-Aware Evaluation for Multi-Turn Medical Consultations** EMNLP 2026
  Youssef Mohamed, **Ahmed Heakl**, Qinrong Cui, Junhong Liang, …, Xiuying Chen [Paper] [Code]
- **Cultural Cues Expose the Financial Knowledge Gap They Were Meant to Close** EMNLP 2026 Findings
  Rania Elbadry, **Ahmed Heakl**, Saeed Almheiri, Fan Zhang, …, Zhuohan Xie [Paper]
- **CEPO: RLVR Self-Distillation using Contrastive Evidence Policy Optimization** Under review
  Ahmed Heakl, Abdelrahman M. Shaker, Youssef Mohamed, Rania Elbadry, …, Salman Khan [Paper] [Code]

Earlier: ArzEn-LLM and ResumeAtlas (Procedia CS 2024), AraSpider (NLPBD 2024), music genre classification (IJCNN 2022), and three applied vision papers (ICINCO 2024, IEEE IMSA 2024).

Honors and Service

- Top Reviewer, NeurIPS 2026. Reviewer for ICLR 2026, NeurIPS 2026, ICLR 2027, and NAACL 2027
- Master's Outstanding Research Award, MBZUAI, 2026. 3rd place nationally in ICPC, Egypt
- Open-source models and data with **1.6M+ downloads**, including AIN (1.5M+), LlamaV-o1 (27k+), and CAMEL-Bench (29k+)
- Fully funded MSc scholarship (MBZUAI) and exchange year at Kyushu University, Japan (selected from about 1,200 applicants)

Skills

- **Languages:** Python, C/C++, CUDA, Java, SQL
- **Machine learning:** PyTorch, deep learning, large language models (LLMs), vision-language models (VLMs), diffusion models, object detection, OCR, speech translation
- **Training and inference:** pretraining, reinforcement learning and RL post-training (GRPO, DPO, RLHF), efficient and on-device inference, multi-node DDP, FSDP, DeepSpeed, Slurm, Docker, AWS, GCP